



Statistics of Financial Markets

Chapter 12: Crypto and Stablecoins



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Outline

Motivation

Cryptocurrency Markets

Bitcoin and Altcoins

Stablecoins

Statistical Properties of Crypto Returns

Applications

Conclusions

References

Motivation

- The emergence of digital assets as a new asset class
- Statistical challenges posed by cryptocurrency markets
- Relevance for portfolio diversification and risk management

Cryptocurrency Markets

- Overview of the cryptocurrency ecosystem
- Market microstructure: exchanges, order books, liquidity
- 24/7 trading and its implications for volatility modeling
- Market capitalization and trading volume dynamics

Bitcoin and Altcoins

- Bitcoin as the benchmark cryptocurrency
- Major altcoins: Ethereum, Ripple, Litecoin
- Return and volatility characteristics
- Cross-cryptocurrency correlations and co-movements

Stablecoins

- ▣ Definition and purpose of stablecoins
- ▣ Types: fiat-collateralized (USDT, USDC), crypto-collateralized (DAI), algorithmic
- ▣ Peg stability analysis and de-pegging events
- ▣ Terra/Luna collapse: a case study

Statistical Properties of Crypto Returns

- ▣ Distribution of cryptocurrency returns: extreme heavy tails
- ▣ Volatility clustering and long memory in crypto markets
- ▣ Comparison of stylized facts: crypto vs. traditional assets
- ▣ Bubble detection and regime switching

Applications

- ▣ Volatility modeling for Bitcoin using GARCH-type models
- ▣ Stablecoin peg deviation analysis
- ▣ Portfolio optimization with crypto assets
- ▣ Python implementation for crypto data retrieval and analysis



Conclusions

- ▣ Summary of crypto market characteristics and stablecoin dynamics
- ▣ Challenges for statistical modeling in crypto markets
- ▣ Future directions and regulatory developments

References

- Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.
- Baur, D.G., Hong, K., Lee, A.D. (2018). Bitcoin: Medium of exchange or speculative assets?
- Lyons, R.K., Viswanath-Natraj, G. (2023). What keeps stablecoins stable?